Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
END SEMESTER EXAMINATIONS, DECEMBER 2024

RENEWABLE ENERGY [ELE 4306]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 02 December 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. How are solar cells classified based on the crystal structure of their semiconductor materials? Which one is more efficient? ***(02)***

1B. The following information is provided for a family biogas digester:

The mass of the dry input per day is 12 kg, the biogas/mass of dry input is 0.2, the burner efficiency is 62%, proportion factor is 0.8 and the heat of combustion of methane is 26 MJ/m^3.

Calculate the power output (continuous thermal) available from the digester. ***(04)***

1C. 'Solar cell quality is maximum when the value of fill factor approaches unity'. Justify this statement with suitable expressions and Graphs. ***(04)***

2A. Explain the two main challenges of Solar powered Irrigation systems in India? ***(02)***

2B. Calculate the number of daylight hours (sunshine hours) in Manipal on 1st February & 2nd December. The latitude of Manipal is 13.35° N. ***(04)***

2C. What are the main types of renewable energy sources, and which of these is best suited for India's energy needs? ***(04)***

3A. What are the drawbacks of a single basin tidal power plant, and how can they be addressed? ***(02)***

3B. An ideal PV cell generates 2W at 0.5V under specific environmental conditions. Calculate the resulting power, current, and voltage if the cells are configured in the following ways:
i) PV cells are arranged in a panel with 4 parallel columns, each containing 10 cells in series. ii) Multiple panels are connected to form an array with 2 parallel columns, each containing 4 panels in series. iii) Several arrays are connected to form a solar system with 10 parallel columns, each containing 20 arrays in series. ***(04)***

3C Draw a block diagram of a wind energy conversion system and describe its key features. ***(04)***

4A. Compute the efficiency of the wind energy conversion system, if its power coefficient is 50%, rotor efficiency is 90%, generator efficiency is 86% and gearbox efficiency is 88% . ***(02)***

4B A horizontal axis wind turbine has a diameter of 8m when the wind speed unaffected by the turbine is 12m/s. The turbine rotates at 280rpm and produces 12kW of mechanical power. Find the tip speed ratio and power coefficient. Assume the density of air ρ=1.225kg/m^3 ***(04)***

4C. Determine the required volume of a cow dung-based biogas plant to meet the daily cooking needs of a family of five adults and the lighting requirements of three 100 CP lamps for four hours. Additionally, calculate the number of cows needed to supply the necessary feed for the plant, using standard data values.

a) Only 70% of the cow dung produced can be collected as 30% is lost in fields during grazing.
b) Solid matter content in cow dung is about 18%
c) Biogas required for cooking is about 0.227m^3/person/day
d) Gas required for lighting a 100CP lamp is 0.126m^3/hour
e) The density of slurry is about 1090kg/m^3
f) Cow dung production rate :12kg/day/head
g) Gas Yield: 0.34 m^3/kg of dry matter
h) Retention period: 60 days ***(04)***

5A. What are the advantages and limitations of geothermal energy in different geographical regions? ***(02)***

5B. What are the different types of Gasifiers? Explain the working of any one of these Gasifiers. ***(04)***

5C. Given a solar power input of 200 W/m^2 during daylight hours, calculate the total area of PV cells required to generate sufficient electricity to operate the following loads. Assume a PV cell efficiency of 20%.

a) A desktop computer with a power requirement of 400W
b) An electric geyser with a power requirement of 1kW
c) An air cooler of 100 watts ***(04)***